

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

x	y	$(x - \bar{x})$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x - \bar{x}) = 0$	$\sum (x - \bar{x})^2 = 10$	$\sum (y - \bar{y}) = 0$	$\sum (x - \bar{x})(y - \bar{y}) = 8$

$$\bar{x} = 1, \bar{y} = 2$$

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

$$b_2 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = 0.8 \Rightarrow \text{slope: The average change in } y \text{ for a one-unit increase in } x.$$

$$b_1 = \bar{y} - b_2 \bar{x} = 1.2 \Rightarrow \text{intercept: The predicted value of } y \text{ when } x = 0.$$

c. Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$

and $\sum (x - \bar{x})(y - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

$$\sum_{i=1}^5 x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$$

$$\sum_{i=1}^5 x_i y_i = 3 \cdot 4 + 2 \cdot 2 + 1 \cdot 3 + (-1) \cdot 1 + 0 \cdot 0 = 18$$

$$\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$$

$$\sum x_i^2 - N\bar{x}^2 = 15 - 5 \cdot 1^2 = 10 = \sum (x_i - \bar{x})^2$$

$$\sum (x - \bar{x})(y - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$$

$$\sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8 = \sum (x - \bar{x})(y - \bar{y})$$

d. Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row. Calculate the sample variance of y ,

$s_y^2 = \sum_{i=1}^N (y - \bar{y})^2 / (N-1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x - \bar{x})^2 / (N-1)$, the sample covariance

between x and y , $s_{xy} = \sum_{i=1}^N (y - \bar{y})(x - \bar{x}) / (N-1)$, the sample correlation between x and y , r_{xy}

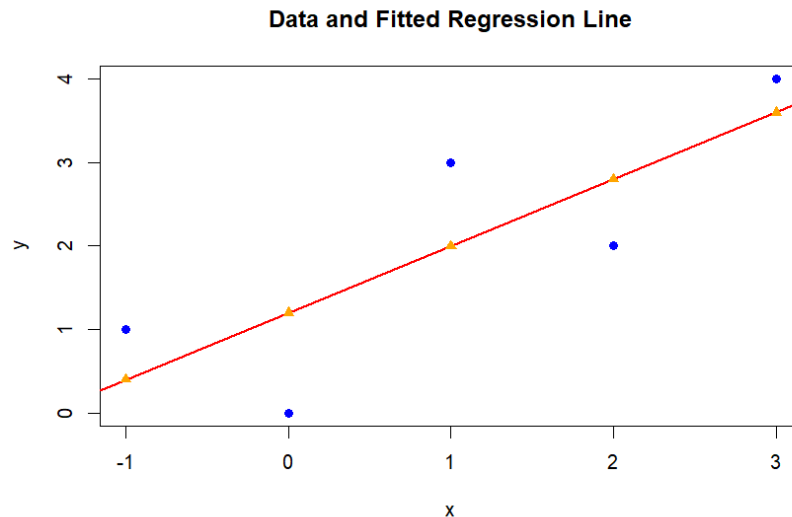
$= s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile,

of x ?

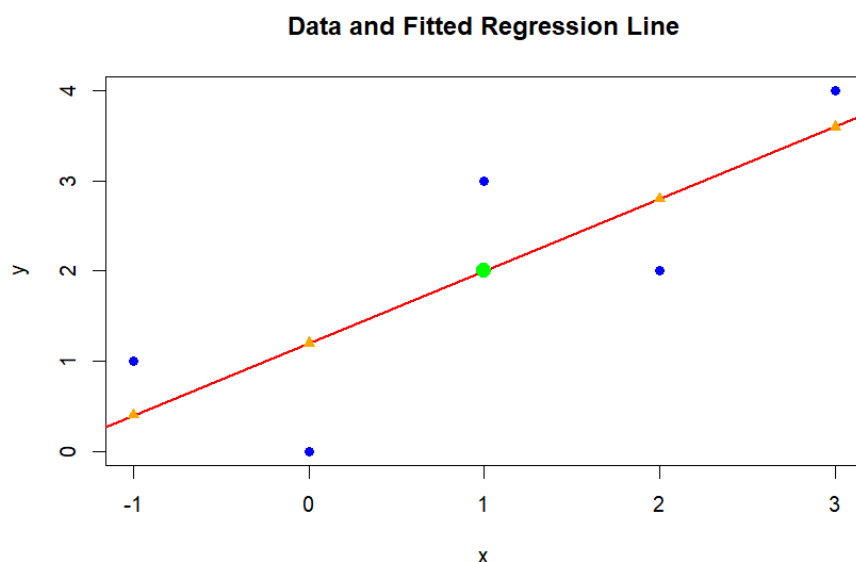
x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2.0	1.0	1.0	1.0
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$s_y^2 = 2.5, s_x^2 = 2.5, s_{xy} = 2, r_{xy} = 0.8, CV_x = 158.1139, \text{ the median, 50th percentile, of } x = 1$$

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2x_i$



f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.



g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

$$b_1 + b_2\bar{x} = 1.2 + 0.8 \cdot 1 = 2 = \bar{y}$$

h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i / N$.

$$\bar{\hat{y}} = \sum \hat{y}_i / N = 10 / 5 = 2 = \bar{y}$$

i. Compute $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \sum \hat{e}_i^2 / (N-2) = 3.6 / 3 = 1.2$$

j. Compute $\widehat{var}(b_2|x)$ and $se(b_2)$.

$$\widehat{var}(b_2|x) = \hat{\sigma}^2 / \sum (x_i - \bar{x})^2 = 1.2 / 10 = 0.12$$

$$se(b_2) = \sqrt{\widehat{var}(b_2|x)} = \sqrt{0.12} \approx 0.34641$$